CIVITAS
Sustainable and smart mobility for all
2030

CIVITAS indicators

Share of households above transport affordability threshold by income group (SOC_EQ_AF1)

DOMAIN

Transport	Environment	Energy	**Society**	Economy

TOPIC	**Equity**	
IMPACT	**Affordability of transport services** *Improving the affordability of transport services*	**SOC_EQ**

Category

Key indicator	Supplementary indicator	State indicator

CONTEXT AND RELEVANCE

Transport affordability refers to the share of household income spent on transport. Spending more than 10% of income on transport is considered a financial burden, especially for lower-income households (European Commission, 2025). Ensuring affordable transport enables people to access essential services, employment, and social activities without undue financial strain, helping to prevent social and economic exclusion. This indicator therefore captures the inclusivity of urban transport systems.

This indicator provides a measure of the affordability of transport in the pilot area. **It is a relevant indicator when the policy action is aimed at reducing travel costs, expanding access to cost-efficient modes, or providing targeted subsidies and compensatory measures to lower income groups. A successful action is reflected in a LOWER value of the indicator.**

DESCRIPTION

This indicator is the **share of households whose transport expenditure is above 10% of income, disaggregated by income quintile**. It is expressed as a vector of five values, each representing the proportion of households exceeding this threshold within a given income quintile. This captures distributional effects and highlights inequalities in how transport affordability affects different income groups.

METHOD OF CALCULATION AND INPUTS

The indicator is calculated using survey responses from a sample of citizens reporting their monthly household income and transport spending. **The indicator should be computed by the user** and its values entered into the supporting tool.

Method	
Based on responses collected by sample survey	Significance: **0.50**

INPUTS

The following information is needed to compute the indicator:

a) **Responses from a sample of residents and users of the pilot area to survey questions regarding their monthly household income and transport expenditure**.
b) **Income quintiles for the pilot area.** These can be sourced from national statistical offices. If data at the exact pilot scale are unavailable, use the closest higher-level geography (e.g. city, metropolitan area, or region).

A suggested formulation of the survey questions is provided in the Guidelines for surveys which are part of the CIVITAS Evaluation Framework and are reported below.

> Q1) How much money do you estimate your household spends on average in one month on urban mobility, considering all the modes of transport you use (e.g., public transport tickets, fuel, parking)?
>
> Average monthly expenditure: _____

Q2) How much money does your household earn on average in one month?

Average monthly income: _____

The experiment would be reflected by a change in the respondents' transport expenditure.

METHOD OF CALCULATION

After collecting survey responses on monthly income and transport expenditure, the indicator should be computed by the user as follows:

- For each respondent, determine whether monthly transport expenditure exceeds 10% of monthly income.
- Group respondents into income quintiles based on their reported income.
- For each income quintile, calculate the proportion of respondents whose expenditure exceeds the 10% threshold relative to the total number of respondents in that quintile.

EQUATIONS

The indicator is computed as follows:

$$\boldsymbol{TrAffIndex_s = \frac{1}{N_s}\sum_{i=1}^{N_s} I_{i,s}}$$

Where:
$I_{i,s} = 1$ if respondent i in income quintile s reports their household spends more than 10% of their monthly income on transport, and $I_{i,s} = 0$ otherwise.

N_s= total number of survey respondents in income quintile s.

ALTERNATIVE INDICATORS

This indicator measures transport affordability as the share of households in the pilot area whose transport expenditure exceeds 10% of income. It requires conducting a sample survey of residents and users of the pilot area to collect data on household income and transport expenditure.

By contrast, **SOC_EQ_AF2** relies on census data on average income in the pilot area combined with a theoretical estimate of public transport expenditure derived from the prices of tickets and passes. **SOC_EQ_AF3** extends this approach by also accounting for the cost of shared mobility services. These two indicators are preferable to assess whole city pilots focusing primarily on the affordability of public transport and/or shared mobility if conducting a survey is burdensome or not feasible. However, these indicators do not capture the effects of measures targeting affordability across a wider range of transport modes, nor those of smaller-scale local pilots; in such cases, SOC_EQ_AF1 should be used instead.